PROJECT ZIA // OS | 100% LOCAL-FIRST | SNAPDRAGON 888 NODE | 0% CLOUD STORAGE

Project Zia: Ambient Home OS

High-efficiency 24/7 autonomous AI companion system powered by a headless Sony Xperia 1 III server node, controlling local IoT relays, LifeDesk SQLite databases, wireless ADB intent bridges, and multi-mode client visualizers.

1 Hardware Blueprint & Asymmetric Edge Topology

Central Server Node: Sony Xperia 1 III

24/7 Headless Daemon

Acts as the 100% heavy-lifting edge server running inside Android Termux with persistent `termux-wake-lock`. Manages Gemini Flash router, subagent pool, local STT/TTS speech loops, and all SQLite bridges.

PROCESSOR	SYSTEM RAM	SUBAGENTS CAP	ENDPOINTS
Snapdragon 888	12 GB LPDDR5	≤ 2 Workers	0.0.0.0:5000 / 8765

MacBook Pro / PC Client

`JSON-RPC 2.0`

Zero-VRAM interface running pure terminal REPL (`cli.py`) or minimal animated grey eye UI. Executes system-level commands, screen locks, and volume triggers remotely.

Android Bedroom Projector

`Wireless ADB :5555`

Automated media streaming bridge. Zia issues remote Android intents directly over Wi-Fi to launch browser sessions pointing to `miruro.tv` without touching the remote.

Android Family Tablet

`ADB VoIP Intent`

Hands-free family communications station. Dispatches wake intents (`KEYCODE_WAKEUP 224`) and launches Botim VoIP call intents on voice demand.

Family Samsung Phones

`Local Webhook Sink`

Automated background notification queue. Captures forwarded SMS and banking verifications, extracting 4-8 digit OTP codes locally with zero cloud exposure.

2 Tri-Tier Visual Interface Spectrum

Minimalist Grey UI

0% VRAM | <15MB RAM

Interactive HTML/CSS animated eyes with mouse cursor tracking, natural double-blinking, and hysteresis audio waveform mouth.

Anime Sprite Companion

0% VRAM | Active on Minimize

Shoulder-level transparent cel-shaded desktop pet with 21 frames, 8 distinct emotion poses, and organic multi-frequency breathing.

Live2D Cubism WebGL

Loaded Strictly On Demand

Full 4K WebGL avatar summoned via `"load avatar"` and completely unloaded via `"stop avatar"` to reclaim 100% GPU memory.

✓ **Asymmetric Resource Efficiency:** All heavy computing (local Vosk STT, neural Edge-TTS synthesis, LifeDesk SQLite aggregation, and Telegram gateway) runs strictly on the Xperia phone server. Client devices run at **<0.5% CPU load**.

3 Step-by-Step Installation & Deployment Flow

01

Preparing the Sony Xperia 1 III Server Node

`Termux / Android`

Install Termux and Termux:API from F-Droid. Enable persistent wake-lock to prevent Snapdragon 888 deep sleep, and set Sony Battery Care limit to 80% to preserve battery lifespan during 24/7 plugged operation.

```
termux-wake-lock
pkg update && pkg install python git termux-api sqlite clang
git clone https://github.com/your-repo/project-zia.git && cd project-zia
pip install -r requirements.txt && python server.py
```

02

Linking LifeDesk SQLite Database (Habits & Finance)

`Local Database`

Connect Zia's backend directly to your existing LifeDesk SQLite databases (`habits.db` and `finance.db`). Enables sub-millisecond, zero-cloud natural language habit logging and expense aggregation.

```
# Place existing databases in data/lifedesk/ or point .env path
LIFEDESK_HABITS_PATH=data/lifedesk/habits.db
LIFEDESK_FINANCE_PATH=data/lifedesk/finance.db
```

03

Connecting Smart Home IoT & Wireless ADB Bridges

`ESP8266 / ADB`

Configure local IP addresses for your ESP8266/NodeMCU relays (Tasmota/Shelly/WLED) and authorize wireless ADB pairing (port 5555) on your bedroom Android projector and family tablet.

```
adb connect 192.168.1.65:5555 # Android Projector
adb connect 192.168.1.75:5555 # Android Tablet
IOT_DEVICE_IP=192.168.1.50 # ESP8266 Desk Lamp Relay
```

04

Remote Telegram Gateway & Neural TTS Voice Notes

`Away-From-Home`

When away from your local Wi-Fi, the bi-directional Telegram gateway allows you to send commands, query prayer schedules, and receive neural Edge-TTS voice note responses directly inside your personal Telegram chat.

```
TELEGRAM_BOT_TOKEN="your_bot_token"
TELEGRAM_ALLOWED_USER_ID="your_telegram_id"
```

05

Agent-Browser CDP Automation & Family OTP Sink

`Web Automation`

Enables token-efficient accessibility tree parsing (`@e1`, `@e2` tokens under 50 tokens) for web actions, and opens a secure local webhook (`POST /api/webhook/notification`) to capture family OTPs automatically.

```
# Query latest OTP via voice or terminal:
zia> what is the latest otp
# "Latest OTP is 849201 from VK-SBIINB (12s ago)."
```

⚠ **Thermal & Memory Sentinel:** Always keep subagent concurrency capped strictly to `max_workers=2` in `engine/subagents.py` to prevent thermal throttling on the Snapdragon 888 chip.

4 Natural Voice & Text Command Cheatsheet

All queries matching local command patterns bypass the cloud LLM entirely, executing in sub-5ms with **0 tokens consumed**. Complex queries seamlessly route to Gemini Flash with persistent memory context.

SPOKEN QUERY / VOICE COMMAND	TARGET SUBSYSTEM	ACTION PERFORMED & HARDWARE BRIDGE	TOKENS
`"Turn on the desk light"`	Local IoT Relay	Dispatches HTTP GET to ESP8266 relay node (Tasmota/Shelly).	0 TOKENS
`"Play Black Clover on projector"`	Android Projector	Fires wireless ADB intent to stream `miruro.tv` over Wi-Fi.	0 TOKENS
`"Call Dad on Botim on my tab"`	Android Tablet	Sends wake event and triggers Botim VoIP calling intent.	0 TOKENS
`"Mark workout and reading done"`	LifeDesk Habits	Updates `habits.db`, calculates daily streak & milestones.	0 TOKENS
`"Log 150 rupees for lunch"`	LifeDesk Finance	Appends transaction to `finance.db` under category 'food'.	0 TOKENS
`"What is the latest OTP?"`	Family Notification Sink	Pulls recent SMS OTP forwarded from Samsung phone queue.	0 TOKENS
`"Next prayer time"`	Palakkad Astronomical	Calculates exact prayer countdown (Lat: 10.7867° N, Long: 76.6548° E).	0 TOKENS
`"Set timer for 15 minutes"`	Alarm Scheduler	Enqueues countdown timer with audio chime alerts.	0 TOKENS
`"Load avatar" / "Stop avatar"`	Dual-Mode Visualizer	Mounts 4K Live2D WebGL or unloads instantly to 0% VRAM.	0 TOKENS
`"Check train schedule to CBE"`	Agent-Browser A11y	Captures compact accessibility tree snapshot of live schedule.	0 TOKENS
`"How should I structure my day?"`	Gemini Flash Brain	Generates natural response with persistent LifeDesk & milestone memory.	LLM TOKEN
`"What's on my screen right now?"`	Multimodal Vision	Captures active desktop window and analyzes error/code via Gemini.	LLM TOKEN

5 Health Check & Diagnostic Commands

Run Full Test Suite:
`python tests/ultimate_test_suite.py`
Validates all 64 subroutines across cross-OS engine & bridges.

Xperia Node Readiness:
`python tests/test_xperia_node_ready.py`
Tests live Gemini LLM latency, dynamic `.env` reload & endpoints.

Project Zia // Lead Systems Architect & Principal Engineer Documentation.
Local-First Autonomous Artificial Intelligence Node. Palakkad, Kerala.